

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	13 July 2026
Team ID	
Project Name	Visualization Tool for Electric Vehicle Charge and Range Analysis
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Data Import	Import EV Population Dataset, EV Specifications Dataset, EV Charging Stations Dataset, and Global EV Sales Dataset into Tableau.
FR-2	Data Preprocessing	Remove duplicate records, handle missing values, change data types, create relationships between datasets, and prepare data for visualization.
FR-3	Dashboard Visualization	Create interactive dashboards using KPI Cards, Bar Charts, Pie Charts, Line Charts, Scatter Plots, Tree Maps, Heat Maps, and Maps to analyze EV data.
FR-4	Interactive Analysis	Enable filters for Brand, Model, Vehicle Type, Model Year, Electric Range, Battery Capacity, Charging Time, and Charging Network.
FR-5	Story & Insights	Develop Tableau Story points to present project overview, EV trends, brand comparison, charging analysis, and final insights.

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The dashboard should have a simple, user-friendly, and interactive interface that allows users to analyze EV data easily.
NFR-2	Security	The dashboard should load quickly and respond efficiently to filters, parameters, and user interactions.
NFR-3	Reliability	The system should provide accurate visualizations based on the imported datasets without data inconsistencies.

NFR-4	Performance	The project should ensure that the datasets remain unaltered and only authorized users can modify the Tableau workbook.
NFR-5	Availability	The dashboard should be accessible whenever the Tableau workbook is opened or published on Tableau Public/Server.
NFR-6	Scalability	The solution should support adding new EV datasets, charging station data, and future dashboards without major modifications.